UNIVERSITY OF THE PEOPLE

Syllabus for Operating Systems 1

Credits: 3
CS 2301

Instructor Contact Information:

You can always send your instructor a private message through the Brightspace Messaging system, accessible via the envelope (Messages) icon in the top navigation bar. Once logged into your course, click your instructor's profile page to see all the ways you can communicate with them, including their email address.

Course Description

The course provides an exploration of operating systems, encompassing key concepts and functionalities. It covers the history, structure, and services of operating systems, progresses through process management including processes, threads, CPU scheduling, and context switching—and delves into concurrency issues such as synchronization and deadlocks. The course also examines input/output management, device drivers, protection mechanisms, and security threats offering a thorough understanding of operating system principles and their evolution.

Learning Objectives

Program Learning Outcomes (PLOs)

1. Students will be able to Provides a framework in applying strategies for the effective design of computing systems.
2. Students will be able to explain apply appropriate methods in the planning, development, and management of design projects.
3. Students will be able to covers the knowledge and skills needed to analyze problems from multiple perspectives and seek resolution through multiple methods and tools.
4. Students will be able to apply mathematics methods effectively to analyze and resolve problems.
5. Students will be able to communicate effectively using well organized arguments and credible supporting evidence.
6. Students will be able to enhance a command of critical thinking with respect to computer ethics, privacy, and security.

Course Learning Outcomes (CLOs)

1. Discuss the core components, services of operating systems, including their historical development and recent trends.
2. Explain process management, scheduling, context switching and concurrency.
3. Apply memory management, virtual memory, Secondary Storage and file system techniques using shell scripting.
4. Discuss I/O management, device drivers, and security mechanisms.

Co/Prerequisites

None.

Course Materials

UoPeople courses use open educational resources (OER) and other materials specifically donated to the University with free permissions for educational use. Therefore, students are not required to purchase any textbooks or sign up for any websites that have a cost associated with them. The main required textbooks for this course are listed below and can be readily accessed using the provided links. There may be additional required/recommended readings, supplemental materials, or other resources and websites necessary for lessons; these will be provided for you in the course's General Information and Forums area, and throughout the term via the weekly course Unit areas and the Learning Guides.

1. Arpaci-Dusseau, R. H., & Arpaci-Dusseau, A. C. (2018). *Operating systems: three easy pieces (1.01 ed.)*. Arpaci-Dusseau Books. Open access through https://pages.cs.wisc.edu/~remzi/OSTEP/
2. Hailperin, M. (2019). *Operating systems and middleware: Supporting controlled interaction.* Thomson Learning, Inc.: San Francisco, CA. (Licensed under CC). Access through https://gustavus.edu/academics/departments/mathematics-computer-science-and-statistics/max/os-book/ or download the PDF.
3. Shotts, W. (2019). *The Linux command line. No Starch Press (5th ed.)*. (Licensed under CC). Access through https://linuxcommand.org/tlcl.php or download the PDF.

Technology Requirements

To fully engage in this course and make the most of your learning,

- **PC Required**: A PC is recommended to ensure full compatibility with course tools and activities.
- **Internet Connection**: A stable internet connection is advised for seamless access to course materials and activities.
- **Browser**: For optimal performance and compatibility with course tools, please use an up-to-date browser.

Students must also have the ability to save documents and files. Typically, University of the People learning materials are provided to students in either Adobe PDF or Microsoft Office compatible formats. Therefore, students need to be able to open and save documents in these formats as well.

Campus Tech Support Email for English programs: support@uopeople.edu

Exam Technical Requirements & Support

To ensure maximum compatibility and reliability, it is highly recommended that you use the most recent version of Google Chrome or Mozilla Firefox when taking your exam. Furthermore, you should test the computer and browser ahead of time and ensure it has no technical issues that might hinder you during the exam period.

If you experience a technical problem during the exam, please contact your instructor immediately.

Proctored Final Exams

The Final Exam will be available from Thursday through Sunday of Week/Unit 9, following the completion of eight units of coursework. The format of the Final Exam is similar to that of the quizzes and may include a combination of question types. You will have only one attempt to complete the exam, and it will be graded electronically.

This Final Exam is proctored using Rosalyn, a secure online proctoring system. In addition to reviewing the Rosalyn Proctoring Exam Guide.pdf, you are responsible for understanding and completing all required proctoring steps.

If you experience technical difficulties:

- Use the live in-app chat within Rosalyn for immediate assistance.
- You may also visit the Rosalyn Help Center for additional help.
- Notify your instructor and support if needed after the session. Support at University of the People

Regular and Substantive Interaction

As your instructor, they will interact and engage with each of you on a regular basis throughout the term to support your learning. They will provide direct instruction related to the course's learning objectives, respond to your questions, grade and/or provide feedback on your submitted coursework, post regular announcements, and engage in the course discussion areas regarding academic course content when appropriate.

Course Expectations and Learning Activities

Discussions

Some units in this course require that you complete a Discussion Assignment. You are required to develop and post a substantive response to the Discussion Assignment in the Discussion Forum. A substantive response is one that fully answers the question that has been posed by the instructor. In addition, you must extend the discussion by responding to at least two (2) of your peers' postings in the Discussion Forum. Grading rubrics are provided in the Discussion Forum for each week. Discussion Forums are only active for each current and relevant learning week, so it is not possible to contribute to the forum once the learning week has come to an end.

Assignments Activities

The assignment activities are graded by your instructor. The grading rubric is listed under the assignment instructions. The grading rubric is a document that outlines the criteria that your instructor will use to grade your work.

Class Introductions

This section is your opportunity to introduce yourself to your classmates and create a vibrant learning community. By sharing your background, interests, and goals, you can create meaningful connections and discover commonalities with your peers.

Quizzes

This course will contain three types of quizzes – the Self-Quiz, the Graded Quiz, and the Review Quiz.

These quizzes may contain multiple-choice, true/false, or short answer questions. The results of the Self-Quiz will not count towards your final grade. However, it is highly recommended that you complete the Self-Quiz to ensure that you have adequately understood the course materials. Along with the Reading Assignments, the results of the Self-Quiz should be used as part of an iterative learning process, to thoroughly cover and test your understanding of course material. You should use the results of your Self-Quiz as a guide to go back and review relevant sections of the Reading Assignments. Likewise, the Review Quiz will not count towards your final grade but should also be used to assist you in a comprehensive review and full understanding of all course material, in preparation for your Final Exam. Lastly, the results of the Graded Quiz will count towards your final grade. Specific instructions on the format and content of the Graded Quiz will be provided by your instructor.

Final Exam

The Final Exam will take place during the Thursday and Sunday of Week/Unit 9, following the completion of eight units of work. The format of the Final Exam is similar to that of the quizzes and may contain a combination of different question types. You will have one attempt to take the exam, and it will be graded electronically. Specific instructions on how to prepare for and take the Final Exam will be provided during Week 8 (located inside the Unit 9 Learning Guide). Final Exams must be taken without the use of course learning materials (both those inside and outside the course). If particular materials are allowed for use during the exam, these will be noted in the exam's instructions.

The Final Exam for this course must be done under the supervision of a proctor. Since you already secured your proctor before registering for this course, this is a reminder that you should coordinate with him/her before you take the exam. As a reminder, students are required to successfully complete proctored exams spaced throughout their program of study at UoPeople, in order to verify the student's identity in confirming a degree and diploma upon graduation.

Course Forum

The Course Forum is the place to raise issues and questions relating to the course. It is regularly monitored by the instructors and is a good place to meet fellow students taking the same course. While it is not required to participate in the Course Forum, it is highly recommended.

Participation Expectations

- Be involved and active in your courses.
- Be highly motivated and disciplined.
- Check the course homepage, calendar and assignment page, the course syllabus, your UoPeople email, and the Brightspace course discussion forums several times a week.
- Post the required comments and responses to the discussion forum for your course.
- Keep up with your assignments and online quizzes/exams (as applicable) and manage your time well. These quizzes test your knowledge and comprehension of the new content.
- Participate actively in class discussions.
- Be polite and respectful.
- Use good grammar and correct spelling.
- Be honest and original. Plagiarism will not be tolerated in any online course.

Non-participation is characterized by lack of any assignment submissions, inadequate contributions to the Discussion Forums, and/or lack of peer feedback to Discussion/Written Assignments. Also, please note the following important points about course participation:

- Assignments must be submitted on or before the specified deadline. A course timeline is provided in the course schedule, and the instructor will specify deadlines for each assignment.
- Occasionally there may be a legitimate reason for submitting an assignment late. Most of the time, late assignments will not be accepted and there will be no make-up assignments.
- All students are obligated to inform their instructor in advance of any known absences which may result in their non-participation.

Feedback and Suggestions

We value your input and would encourage you to complete the end of course survey to provide us with course feedback and suggestions, and report issues.

Evaluation and Grading Scale

Grading Weights:

Category	% Of Grade	Grade Items (Learning Activities)	Associated Learning Objectives/Outcomes
Discussion Forums	30%	1. Unit 1 – Discussion Forum 2. Unit 3 – Discussion Forum 3. Unit 4 – Discussion Forum 4. Unit 5 – Discussion Forum 5. Unit 6 – Discussion Forum 6. Unit 8 – Discussion Forum	• Unit 1 – CLO 1 • Unit 3 – CLO 2 • Unit 4 – CLO 3 • Unit 5 – CLO 3 • Unit 6 – CLO 4 • Unit 8 – CLO 1
Assignment Activities	40%	1. Unit 2 – Assignment Activity 2. Unit 3 – Assignment Activity 3. Unit 4 – Assignment Activity 4. Unit 5 – Assignment Activity 5. Unit 7 – Assignment Activity	• Unit 2 – CLO 2 • Unit 3 – CLO 2 • Unit 4 – CLO 3 • Unit 5 – CLO 3 • Unit 7 – CLO 4
Graded Quiz	10%	1. Unit 3 – Graded Quiz 2. Unit 6 – Graded Quiz	• Unit 3 – CLO 2 • Unit 6 – CLO 4
Final Exam	20%	1. Unit 9 – Final Exam	• Unit 9 – CLO 1, 2, 3, 4
TOTAL	**100%**		

Grading Scale:

Letter Grade	% Grade	Grade Points
A	98%-100%	4.00
A	93-97%	4.00
A-	90%-92%	3.67
B+	88%-89%	3.33
B	83%-87%	3.00
B-	80%-82%	2.67
C+	78%-79%	2.33
C	73%-77%	2.00
C-	70%-72%	1.67
D+	68-69%	1.33
D	63%-67%	1.00
D-	60%-62%	0.67
F	<60	0.00
W	N/A	N/A

Students may also be granted Withdrawal (W), if they withdraw from the course, or an Incomplete (I) should their circumstances permit.
A student who feels they were graded unfairly, or who seeks to dispute a grade, may initiate a grade appeal process. Refer to University Policies for more information on withdrawals and appeals.

Course Schedule

UNIT 1: Introduction to Operating Systems

- Watch/Read the reading assignments due 3rd – 4th day of the week (recommended)
- Discussion first response due 4th day of the week
- Discussion replies due 6th day of the week
- Self-quiz due 7th day of the week

UNIT 2: Process Management & CPU Scheduling

- Watch/Read the reading assignments due 3rd – 4th day of the week (recommended)
- Assignment activity due 6th day of the week
- Self-quiz due 7th day of the week

UNIT 3: Concurrency

- Watch/Read the reading assignments due 3rd – 4th day of the week (recommended)
- Discussion first response due 4th day of the week
- Discussion replies due 6th day of the week
- Assignment activity due 6th day of the week
- Self-quiz due 7th day of the week
- Graded quiz due 7th day of the week

UNIT 4: Memory Management

- Watch/Read the reading assignments due 3rd – 4th day of the week (recommended)
- Discussion first response due 4th day of the week
- Discussion replies due 6th day of the week
- Assignment activity due 6th day of the week
- Self-quiz due 7th day of the week

UNIT 5: Storage Management and Shell Scripting

- Watch/Read the reading assignments due 3rd – 4th day of the week (recommended)
- Discussion first response due 4th day of the week
- Discussion replies due 6th day of the week
- Assignment activity due 6th day of the week
- Self-quiz due 7th day of the week

UNIT 6: Input / Output Management

- Watch/Read the reading assignments due 3rd – 4th day of the week (recommended)
- Discussion first response due 4th day of the week
- Discussion replies due 6th day of the week
- Self-quiz due 7th day of the week
- Graded quiz due 7th day of the week

UNIT 7: Security and Protection

- Watch/Read the reading assignments due 3rd – 4th day of the week (recommended)
- Assignment activity due 6th day of the week
- Self-quiz due 7th day of the week

UNIT 8: Advanced Topics and Case Studies

- Watch/Read the reading assignments due 3rd – 4th day of the week (recommended)
- Discussion first response due 4th day of the week
- Discussion replies due 6th day of the week
- Self-quiz due 7th day of the week
- Review quiz due 7th day of the week

UNIT 9: Final Exam

- Final exam due 7th day of the week

University Policies & Processes

Late Work/Make-up Policy

Please review the Late Work policy in the University Catalog.

Code of Conduct

University of the People expects that students conduct themselves in a respectful, collaborative, and honest manner at all times. Harassment, threatening behavior, or deliberate embarrassment of others will not be

permitted. Any conduct that interferes with the quality of the educational experience is not allowed and may result in disciplinary action, such as course failure, probation, suspension, or dismissal. For more information on this topic, please review the General Code of Conduct in the University Catalog.

Procedures for Resolving Academic Grievances/Appeals

If you believe that the final grade you received for a course is erroneous, unjust, or unfair, please contact your course instructor. This must be done within fourteen days of the last day of the term. For more information on this topic, please review the Grievance Policy and Grade Appeals Procedure in the University Catalog.

Withdrawal and Drop Date Policy

Please review the Course Drops and Withdrawals policy of the University Catalog.

Academic Integrity and Plagiarism

Plagiarism is a serious breach of academic trust. In academic work, our words and ideas are the value of our work, so turning in someone else's work as if it were your own is a form of theft. When you use someone else's words and ideas--whether it's the work of a famous writer or a fellow student--without crediting the source or authorship of those words and ideas, you are plagiarizing.
Please review the Code of Academic Integrity in the University catalog for more information.

Reference: Lee, C. (2019, April 4). *How to address Academic Integrity in your course syllabus.* Turnitin. https://www.turnitin.com/blog/how-to-address-academic-integrity-in-your-course-syllabus

Intellectual Property

UoPeople respects the intellectual property rights of others who seek to create, preserve, and disseminate knowledge through teaching, collective learning, and continued research at the University at large. For more information on this topic, please review the Intellectual Property policy in the University catalog.

Reasonable Accommodations

Contact your Program Advisor to open a request for support.

Student Support Services & Resources

English Programs

Academic Advising: advising@uopeople.edu
Financial Aid: financial.aid@uopeople.edu
Library Resources: library@uopeople.edu
Payment Processing: payments@uopeople.edu
Student Services: student.services@uopeople.edu
Technical Support: support@uopeople.edu